

Introduction to Social Data Analytics: Week 8 Part 2

Today's Lecture

- ▶ Discuss how to handle dates in R using the `lubridate` package
- ▶ Introduce the `ggplot2` package
- ▶ How to customize figures in `ggplot2`

Three types of dates in lubridate

- ▶ Date
 - ▶ Time
 - ▶ Date-time
- ▶ In R, we can use the `lubridate()` package to work with dates

```
library(lubridate)
```

```
## Loading required package: timechange
```

```
##
```

```
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      date, intersect, setdiff, union
```

We can add days, months, years to dates in R

```
ymd("2012-01-22") + hours(5)
```

```
## [1] "2012-01-22 05:00:00 UTC"
```

```
ymd("2012-01-22") + days(1)
```

```
## [1] "2012-01-23"
```

```
ymd("2012-01-22") + months(2)
```

```
## [1] "2012-03-22"
```

```
ymd("2012-01-22") + years(5)
```

```
## [1] "2017-01-22"
```

Application: Air Pollution Monitoring

► Load data:

```
library(haven)
library(tidyverse)
pm <- read_dta("station_day_1116.dta")
```

Date object is from Stata

- Coincides with the number of days after January 1, 1960.

```
head(pm$date)
```

```
## [1] 18628 18629 18630 18631 18632 18633
```

Create R date object

- Coincides with the number of days after January 1, 1960.

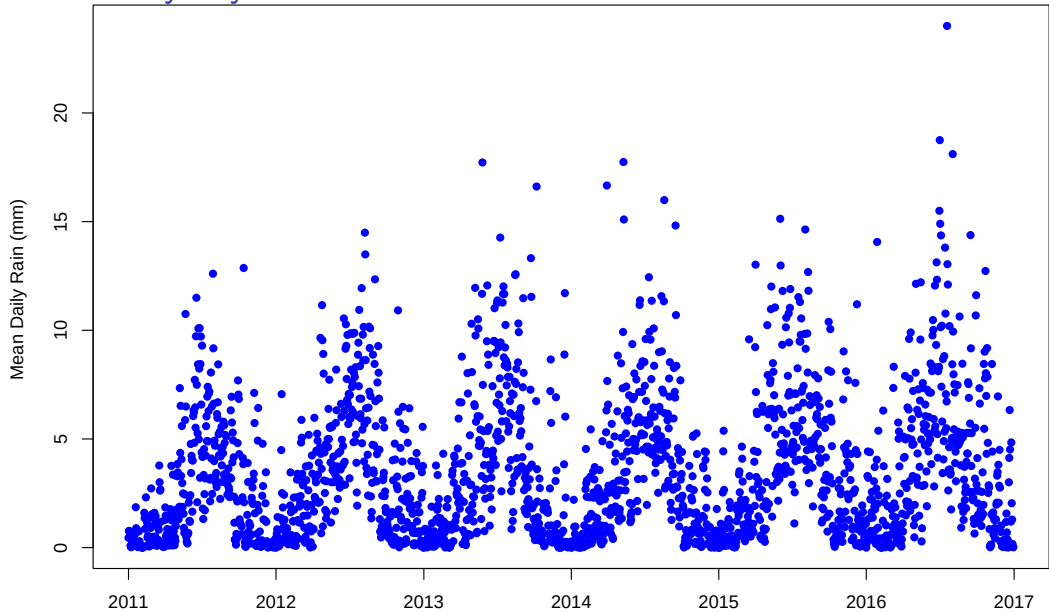
```
head(ymd("1960-01-01") + days(pm$date))
```

```
## [1] "2011-01-01" "2011-01-02" "2011-01-03" "2011-01-04" "2011-01-05"  
## [6] "2011-01-06"
```

Plot rainfall by day

```
pm_byrdate <-  
  pm %>%  
  mutate(rdate = ymd("1960-01-01") + days(pm$date)) %>%  
  group_by(rdate) %>%  
  summarize(meanrain = mean(rain, na.rm=T))  
  
#Plot rain by day  
plot(pm_byrdate$rdate, pm_byrdate$meanrain, col="blue",  
      pch=16, xlab="Date", ylab="Mean Daily Rain (mm) ")
```


Plot rainfall by day



Now: ggplot2

- ▶ ggplot2 is a very popular graphics package in R
- ▶ It will allow you to create all the figures we've learned how to make so far and many many more
 - ▶ Additionally, all plots follow a similar syntax, making it easy to learn how to code new plots quickly
- ▶ Before we get started on ggplot2, we will get our pollution data ready by filtering out missing values and then grouping by time relative to automation

```
pm <- filter(pm, !is.na(pm10))
```

Group air pollution by time to automation

- #1. Caculate time after automation*
- #2. Filter for 1 year before and after*
- #3. Group by day*
- #4. Calculate mean pm10 and mean rain*
- #5. Create a variable to indicate after automation*

```
pm_byday <- pm %>%  
  mutate(  
    T=date-auto_date) %>%  
  filter(T>-364, T<364) %>%  
  group_by(T) %>%  
  summarize(meanpm10 = mean(pm10),  
    meanrain=mean(rain)) %>%  
  mutate(after=T>0)
```

Template for GGPlot2

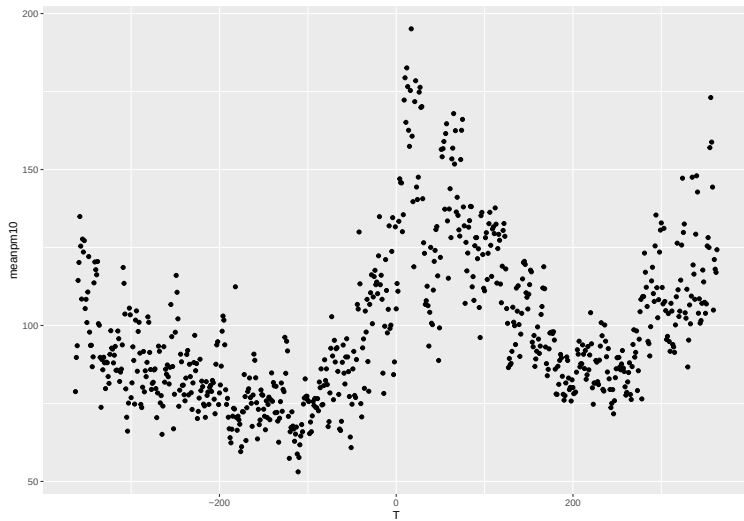
- From Grolemond and Wickham and *R for Data Science*, Part 3:

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>))  
}
```

Create scatterplot in ggplot

```
#Basic scatterplot  
ggplot(data = pm_byday) +  
  geom_point(mapping = aes(x = T, y = meanpm10))
```

Create scatterplot in ggplot



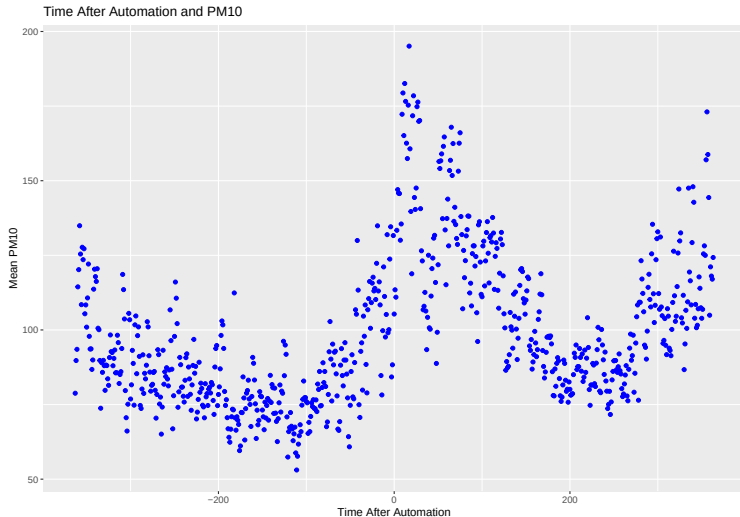
Customizing your ggplot

- ▶ There are many many ways to customize a ggplot
- ▶ We will explore some customization of the scatter plot we made earlier
- ▶ There are a lot of online resources for customizing ggplot2 objects (see R studio graph gallery for one useful resource)

Customize your ggplot

```
ggplot(data = pm_byday) +  
  geom_point(mapping = aes(x = T, y = meanpm10),  
    color="blue") +  
  xlab("Time After Automation") + ylab("Mean PM10") +  
  ggtitle("Time After Automation and PM10")
```

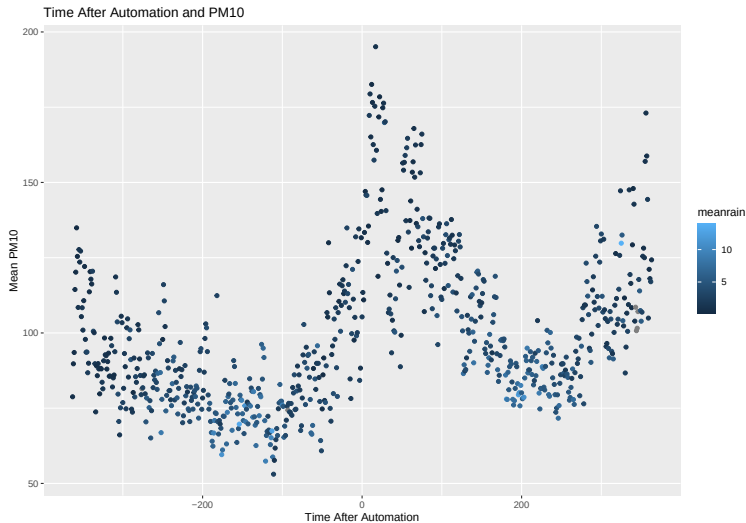

Customize your ggplot



Customize your ggplot by a third variable

```
#By color  
ggplot(data = pm_byday) +  
  geom_point(mapping = aes(x = T, y = meanpm10,  
    color=meanrain)) +  
  xlab("Time After Automation") + ylab("Mean PM10") +  
  ggtitle("Time After Automation and PM10")
```

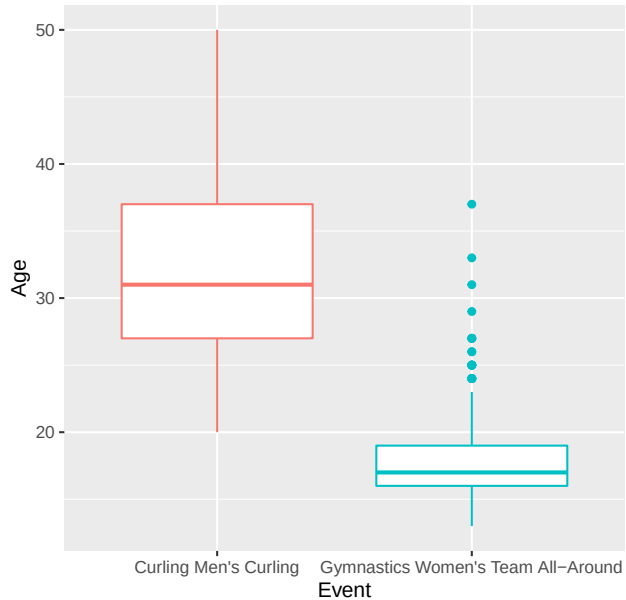
Customize your ggplot by a third variable



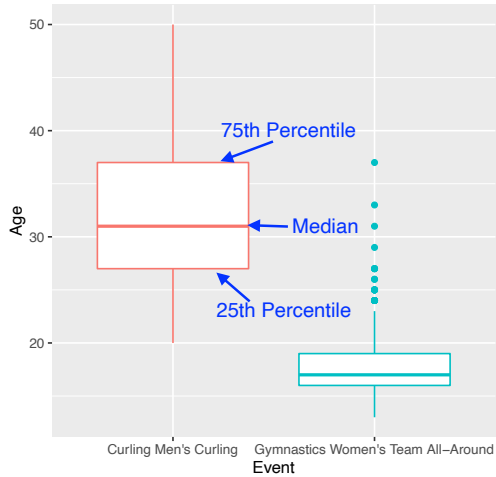
Boxplot Example

- ▶ To understand boxplots theoretically, we are first going to go through an example
- ▶ The data comes from data on the ages of Olympic athletes (which you will use in this week's lab)
 - ▶ We are going to use this data to understand the distribution of ages of athletes across two different events

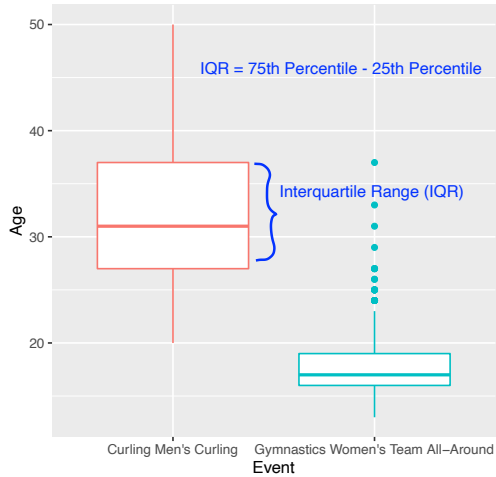
Boxplot of Ages For Different Olympic Sports



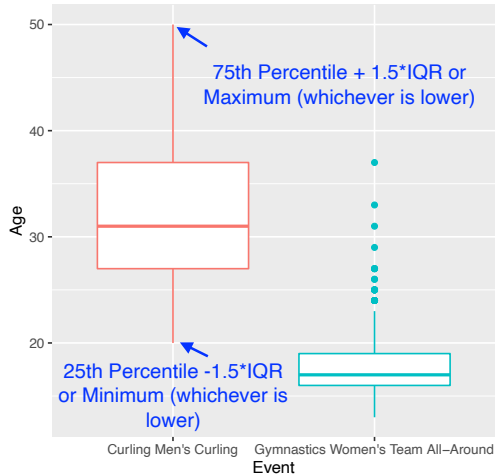
Boxplot of Ages For Different Olympic Sports



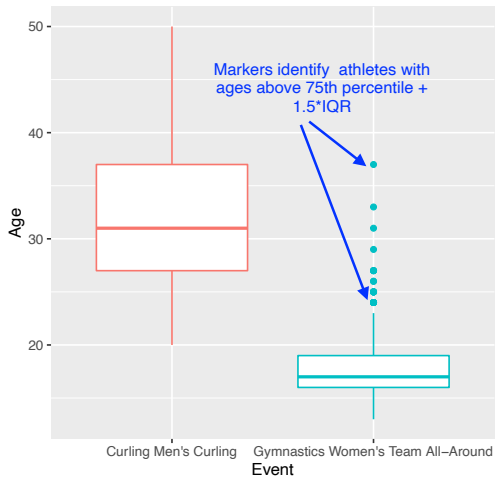
Boxplot of Ages For Different Olympic Sports



Boxplot of Ages For Different Olympic Sports



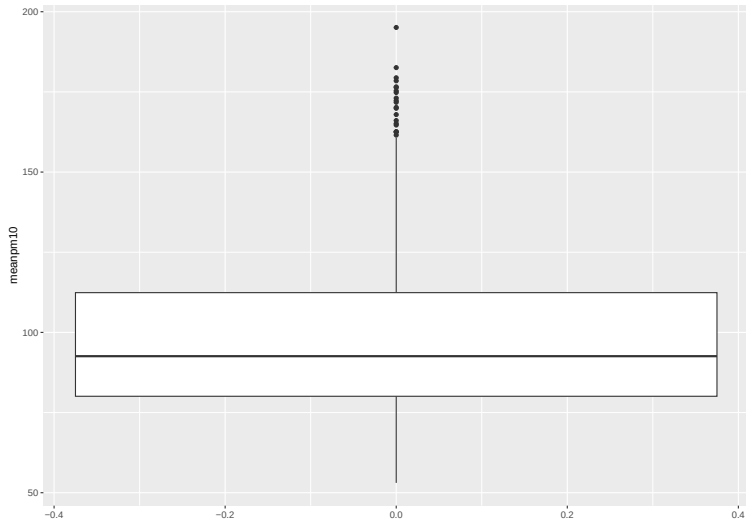
Boxplot of Ages For Different Olympic Sports



Create boxplot in ggplot

```
#Basic boxplot  
ggplot(data = pm_byday) +  
  geom_boxplot(mapping = aes(y = meanpm10))
```

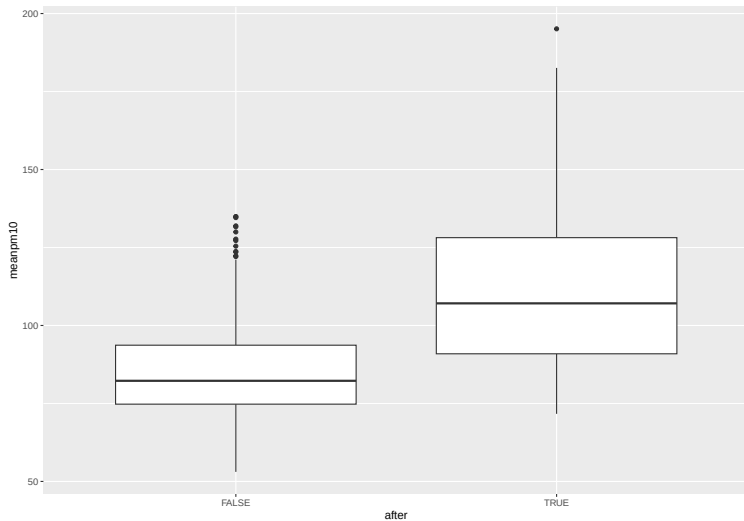
Create boxplot in ggplot



Create boxplot in ggplot by an X

```
#Boxplot by after  
ggplot(data = pm_byday) +  
  geom_boxplot(mapping = aes(x = after, y = meanpm10))
```

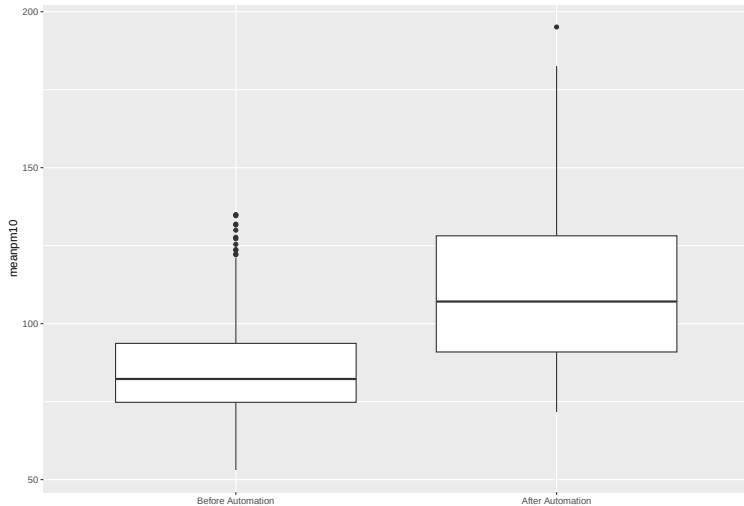
Create boxplot in ggplot by an X



Change the x-axis labels

```
#Boxplot by after  
ggplot(data = pm_byday) +  
  geom_boxplot(mapping = aes(x = after, y = meanpm10)) +  
  scale_x_discrete(labels=c("Before Automation", "After Automation")) +  
  xlab("")
```

Change the x-axis labels



Conclusion

- ▶ Greenstone et al. (2022) finds an increased in reported air pollution after automatic air pollution monitoring systems are introduced
- ▶ This design in which a “treatment” occurs at a threshold is called a regression discontinuity design
 - ▶ It is a very common method used in social science research
- ▶ ggplot2 gives us a way to nicely present the results of Greenstone et al. (2022)